



Development of a GMO-free *de novo* lager yeast with reduced phenolic off-flavor production through interspecific hybridization and UV mutagenesis

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ABSTRACT

Lager beers are traditionally brewed using *Saccharomyces pastorianus*, a natural hybrid between *S. cerevisiae* and *S. eubayanus* that combines robust fermentation with cold tolerance. However, commercial lager yeasts originate from a narrow genetic pool, limiting strain diversity and aromatic complexity. To meet growing demand for new lager beer profiles, the development of novel non-GMO strains is increasingly important. Many wild yeasts from non-brewing environments can ferment wort but produce phenolic off-flavor (POF⁺) by converting ferulic acid (FA) into 4-vinyl guaiacol (4-VG), restricting their industrial applicability. In this study, a *de novo* POF⁺ lager-like hybrid (FM1) generated by spore-to-spore mating between a monosporic culture of the sourdough strain *S. cerevisiae* Y15 and *S. eubayanus* x *S. uvarum* NBRC1948, was submitted to UV mutagenesis to reduce POF formation. Screening on FA-supplemented medium identified a POF-reduced mutant, FM1.45, showing lower ferulic acid conversion and higher cinnamic acid sensitivity than FM1. Wort microfermentation trials demonstrated that FM1 and FM1.45 both outperformed the parental strains in fermentation kinetics and produced similar aroma profiles, with FM1.45 generating substantially less 4-VG than FM1. Sanger sequencing results were consistent with a loss of the *S. eubayanus* *PAD1-FDC1* gene cluster and a frameshift mutation in the *S. cerevisiae* *FDC1* gene, leading to a non-functional ferulic acid decarboxylase. Flow cytometry supported a ploidy increase from 3n to 4n in FM1.45, which likely accounts for the restored spore viability observed in this mutant compared with the sterile FM1. Overall, this work demonstrates that combining interspecific hybridization with UV mutagenesis represents an effective GMO-free strategy for generating novel lager yeast strains with robust fermentation performance and reduced POF production.

1. Introduction

Brewer's yeasts predominantly belong to the genus *Saccharomyces* and are traditionally categorized into ale and lager yeasts (Capece et al., 2018). Ale yeast consists of *Saccharomyces cerevisiae* strains, whereas lager yeast comprises natural hybrids (historically referred to as *Saccharomyces pastorianus*) between *S. cerevisiae* and the cryotolerant species *Saccharomyces eubayanus*. In both groups, brewing yeast genomes exhibit clear signatures of domestication, such as efficient maltose and maltotriose utilization and the loss of phenolic off-flavor

(POF) production (Gallone et al., 2016, 2019; Gonçalves et al., 2016). These traits underpin their adaptation to ferment wort at high and low temperatures, respectively. Lager beer accounts for more than 90% of global beer production, making it the primary focus of brewing research. At the same time, the rapid expansion of craft beer market (Garavaglia and Swinnen, 2018) and the increasing consumer interest in diverse beer styles (Aquilani et al., 2015) have stimulated efforts to develop novel brewing starter cultures. However, introducing novel characteristics into alcoholic beverages with commercial *S. pastorianus* strains is challenging, as they are genetically and phenotypically limited, highlighting

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the demand for improved strains with enhanced technological and sensory properties (Cubillos et al., 2019; Molinet and Cubillos, 2020).

A useful strategy to develop new brewing stater cultures is the exploitation of *S. cerevisiae* from non-brewing niches (Iattici et al., 2020). A major challenge in incorporating these wild yeasts into brewing strain development pipeline, is that these strains lack the domestication signatures present in traditional brewing yeasts. As a result, wild strains often produce POF compounds during fermentation. These off-flavors are formed from phenolic acids such as hydroxycinnamic and hydroxybenzoic acids derived from wort through yeast enzymatic bioconversion. These include 4-vinyl guaiacol (4-VG) and 4-vinyl phenol produced via decarboxylation of ferulic and coumaric acids, respectively (Coghe et al., 2004). Both metabolites impart clove-like, smoky, or medicinal aromas that are undesirable in most beer styles.

Yeasts able to decarboxylate ferulic acid (FA) into 4-VG are described as POF⁺ (POF⁺). The POF⁺ phenotype is governed by the *PAD1-FDC1* gene cluster, located on the right arm of chromosome IV in *S. cerevisiae*. *FDC1* encodes the ferulic acid decarboxylase, while *PAD1* encodes a prenylated flavin mononucleotide (FMN) synthase-like enzyme required to produce the prFMN cofactor essential for Fdc1 activity (Mukai et al., 2010; Huang et al., 2012; Lin et al., 2015; Richard et al., 2015). During domestication of brewing yeasts, selection strongly favored POF[−] strains harboring unfunctional alleles of one or both genes (Gallone et al., 2016). In contrast, in natural environments the POF⁺ phenotype may confer a selective advantage by enabling the detoxification of lignin-derived phenolic acids, through their conversion into less harmful vinyl derivatives (Clausen et al., 1994; Larsson et al., 2001; Adeboye et al., 2015; Bautista-Guerrero et al., 2022). Consequently, functional *PAD1* and *FDC1* allele are typically retained in wild *S. cerevisiae* populations or in those strains used for baking and bio-ethanol production (Gallone et al., 2016).

Since the discovery of *S. eubayanus* in Patagonia in 2011 (Libkind et al., 2011), the construction of *de novo* hybrids between domesticated *S. cerevisiae* ale strains and wild *S. eubayanus* through spore-to-spore mating, rare mating, or mass mating, has emerged as an effective, genetically modified organisms (GMO)-free strategy for expanding lager strain diversity (Steensels et al., 2014; Mertens et al., 2015; Hebly et al., 2015; Magalhães et al., 2017; Alexander et al., 2016; Krogerus et al., 2016, 2017, 2018; Gorter de Vries et al., 2019). By combining alleles from different parental species, hybrids displayed enhanced fermentation kinetics, improved sugar utilization, higher stress tolerance, and optimized aroma profiles compared with their parents (Gibson et al., 2017; Steensels and Verstrepen, 2014). Consequently, newly generated lager hybrids hold substantial promises for producing beers with diversified and innovative aroma profiles. However, crossing POF[−] *S. cerevisiae* with POF⁺ *S. eubayanus* typically results in POF⁺ hybrids (Baker et al., 2015). To overcome this drawback, several attempts have been made to generate *S. eubayanus* POF[−] strains. For instance, Mertens et al. (2019) successfully exploited CRISPR-Cas technology to inactivate *FDC1*, although the resulting cold-tolerant parental lineage was classified GMO under European legislation. Diderich et al. (2018) applied random mutagenesis to obtain a GMO-free POF[−] *S. eubayanus* derivative suitable as cryotolerant parental lineage for sexual breeding. However, their proof-of-concept *de novo* lager hybrid can be considered GMO due to the marker-assisted mating of the POF[−] *S. eubayanus* derivative with a haploid *ura3::KanMX*, POF[−] *S. cerevisiae* strain. Consumer concerns about use of GMOs for food and beverage production strongly limit the use of these approaches in brewing industry (Nasuti et al., 2025).

In our previous work, we successfully mated a POF⁺ *S. cerevisiae* strain from sourdough with the *S. eubayanus* x *S. uvarum* brewing strain NBRC1948. The resulting hybrid (Y15.2B x NBRC1948, hereafter referred to as FM1) exhibited heterosis, displaying superior fermentation performance compared with their parental strains, and inherited the ability to ferment maltotriose from NBRC1948 (Catallo et al., 2021). Despite the technological advantages of hybridization, preliminary pilot-scale wort fermentations demonstrated the formation of a clove-

like off-flavor associated with 4-VG production, which hindered the exploitation of the FM1 hybrid for lager beer production. In this study, we applied random UV mutagenesis to hybrid FM1 to obtain a GMO-free hybrid derivative with reduced POF activity and diversified flavor properties suitable for lager brewing.

2. Materials and methods

2.1. Chemicals, strains, and culture conditions

All media and chemicals used in this study were purchased from Oxoid (Basingstoke, Hampshire, UK) and Sigma Aldrich (St. Louis, MO, USA), respectively, except where differently indicated.

Strains used in the present work are listed in Table 1. Hybrid strain FM1 was previously constructed by spore-to-spore mating using a micromanipulator and asci partially digested with zymolyase from the parental strains *S. cerevisiae* Y15.2B and *S. eubayanus* × *S. uvarum* NBRC1948 (Catallo et al., 2021). Yeasts were routinely cultured on YPD medium (1% w/v yeast extract, 1% w/v peptone, 2% w/v dextrose) supplemented with 2% w/v agar when required and incubated at 28 °C for 48 h. Strains were subsequently stored at 4 °C for the duration of the experiments. For long-term storage, strains were preserved at −80 °C in YPD medium supplemented with 25% (v/v) glycerol as cryoprotectant.

2.2. UV mutagenesis protocols

A conventional UV mutagenesis was performed as described by Hashimoto et al. (2005) and Li et al. (2023). Firstly, the lethality effect of two UV sources was evaluated using both a haploid (ale11.2B) and a diploid strain (BY4743), respectively. Overnight cultures in YPD (28 °C, 10 mL) were harvested at 0.5 OD₆₀₀, washed, and resuspended in sterile distilled water. Serial dilutions (10⁰–10^{−4}) were plated on YPD. Cells were exposed to UV using either a Hoefer crosslinker (52.5 mJ; 0, 10, 20, 30 s) or a UV lamp (50 cm distance; 0, 5, 15, 30 s). Plates were incubated in the dark (26–27 °C, ≥5 days) to prevent photoreactivation. Following incubation, colony-forming units (CFU)/mL were counted and compared among treatments to determine mortality percentage according to the following formula:

$$\text{Mortality} = 100 - \left(\frac{\text{CFU}_t \times 100}{\text{CFU}_{t_0}} \right) \quad (1)$$

where CFU_t was the CFU value at each time; CFU_{t0} was the CFU value at the starting point. Optimal UV mutagenesis condition was considered with mortality range between 65 and 85% (Hashimoto et al., 2005).

Table 1

List of yeast strains employed in the present study.

Species	Code	Description; source; reference
<i>S. cerevisiae</i>	Y15.2B	monosporic culture of Y15 isolated from sourdough of Maiorca flour; Catallo et al. (2020)
<i>S. eubayanus</i> × <i>S. uvarum</i>	NBRC1948	Brewing environment
<i>S. eubayanus</i> × <i>S. uvarum</i> × <i>S. cerevisiae</i>	FM1	Catallo et al. (2021)
<i>S. cerevisiae</i>	BY4743	diploid <i>MATa/MATα</i> , <i>his3Δ1/his3Δ1</i> , <i>leu2Δ0/leu2Δ0</i> , <i>lys2Δ0/lys2Δ0</i> , <i>ura3Δ0/ura3Δ0</i> strain; Euroscarf
<i>S. cerevisiae</i>	ale11.4B	<i>MATa</i> haploid monosporic culture of ale brewing strain ale11; LYB LAB culture Collection
<i>S. pastorianus</i>	A-63015	VTT Culture Collection
<i>S. pastorianus</i>	JB712	LYB LAB culture Collection
<i>S. cerevisiae</i>	C-6	Diploid control strain; Peter et al. (2018)
<i>S. cerevisiae</i>	YS8(E)	Tetraploid control strain; Peter et al. (2018)

In addition to conventional UV mutagenesis, the protocol was modified by introducing a nystatin treatment to selectively inhibit fast-growing cells, according to Colomer (2019). Yeast cultures grown overnight in YPD were harvested at 2.5 OD₆₀₀, washed with TE buffer, and resuspended in TE buffer supplemented with 0.03% Triton X-100. Cells were exposed to UV lamp (30 s) in a sterile Petri dish, then incubated for 24 h at 27 °C in YPD medium (aluminum-wrapped tubes).

Cultures were diluted to 0.05 OD₆₀₀ in YPD medium (20 mL final volume) supplemented with 1 mM FA, incubated at 25 °C for 48 h under shaking condition (150 rpm), and then transferred into YPD medium supplemented with 2.5 µg/mL nystatin. Following incubation under the same conditions reported above, the cycle was repeated three times to apply selective pressure. Finally, cells were washed, grown in YPD (25 °C, 24 h), serially diluted, and plated on YPD medium.

In both protocols, mutants impaired in FA decarboxylation were screened via replica plating on YPD supplemented with 4 mM FA. Colonies unable to grow in the presence of FA were re-isolated from the master plate, purified, and stored at −80 °C in YPD medium supplemented with 25% (v/v) glycerol for further analyses.

2.3. Determination of ferulic acid conversion and cinnamic acid sensitivity indices

Ferulic acid conversion index (FACI) and cinnamic acid sensitivity (CASI) index were determined as previously reported with a few modifications (Diderich et al., 2018). Specifically, for FACI assay, yeast strains were inoculated at 1×10^6 CFU into 100 mL Erlenmeyer flasks containing 20 mL YPD medium supplemented with FA (1 mM), with YPD without FA used as control. Incubation was carried out for 72 h at 25 °C and 150 rpm. Following incubation, 2 mL of culture were centrifuged at 12,000 ×g for 10 min at 4 °C. Supernatants were collected and absorbance was measured at 310 nm in three technical replicates, using uninoculated medium as blank. Ferulic acid conversion was expressed as the FACI, using the following formula:

$$\text{FACI} = 100 - \left(\frac{\text{Abs}_{310 \text{ nm}} \text{ culture with 1 mM FA}}{\text{Abs}_{310 \text{ nm}} \text{ culture medium with 1 mM FA}} \right) \times 100 \quad (2)$$

In the case of CASI assay, yeast strains were inoculated into 20 mL YPD supplemented with 1 mM cinnamic acid (CA) and into YPD medium as control. Following incubation under the same conditions reported above, cell growth was assessed by measuring the optical density at 660 nm (OD₆₆₀) in both YPD and YPD media supplemented with CA to evaluate growth inhibition using the following formula:

$$\text{CASI} = 100 - \left(\frac{\text{Abs}_{660 \text{ nm}} \text{ culture with 1 mM CA}}{\text{Abs}_{660 \text{ nm}} \text{ culture without CA}} \right) \times 100 \quad (3)$$

2.4. UHPLC/MS analysis

In the case of FACI assay, 10 mL of final samples were centrifuged at 8000 rpm for 10 min under refrigerated conditions. The resulting supernatants were properly diluted and submitted to Q Exactive Hybrid Quadrupole-Orbitrap Mass Spectrometer coupled to a UHPLC Ultimate 3000 system (Thermo Fisher Scientific, San Jose, CA, USA) for quantifying FA. Preliminarily, the samples were injected into a reversed-phase C18 column (Acquity UPLC HSS C18, 2.1 × 100 mm, 1.8 µm, Waters, Milan, Italy). Chromatographic separation was achieved using a binary gradient of 0.1% formic acid in water and 0.1% formic acid in acetonitrile. Mass spectrometry parameters and gradient details were applied as described by Cattivelli et al. (2023). Quantification was carried out using an external calibration curve. YPD and FA-supplemented YPD media without cells were used as controls.

2.5. Growth inhibition assay

Growth inhibition in presence of 4 mM FA was measured in

microplate cultivations. The microcultures were carried out in 100-well honeycomb microtitre plates at 25 °C (with continuous shaking), and their growth dynamics were monitored with a Bioscreen C MBR incubator and plate reader (Oy Growth Curves Ab, Finland). The wells of the microtitre plates were filled with 300 µL of defined medium (0.67% w/v yeast nitrogen base with amino acids, 8% w/v glucose) supplemented with or without 4 mM FA. Precultures of the strains were started in 20-mL YPD medium and incubated at 25 °C with shaking at 120 rpm overnight. The cultures were centrifuged and the yeast pellets were washed once with sterile deionized water. Cell density was determined as OD₆₀₀ using a bench spectrophotometer after appropriate dilution to ensure measurements within the linear range, and the washed cells were resuspended in defined medium to obtain an OD₆₀₀ = 10 inoculum stock. The microcultures were started by inoculating the microtitre plates with 3 µL of cell suspension per well (for an initial OD₆₀₀ value of 0.1) and placing the plates in the Bioscreen C MBR. Optical density of the microcultures at 600 nm was automatically read every 30 min. Four replicates were performed for each strain in each medium. Growth inhibition percentages were calculated after 8, 10, and 12 h according to Eq. (4).

$$\text{Growth inhibition (\%)} = 100 - \left(\frac{\text{OD}_{600\text{FA}}}{\text{OD}_{600\text{CTRL}}} \times 100 \right) \quad (4)$$

where OD_{600FA} was the optical density of the microcultures at 600 nm in the presence of 4 mM FA; OD_{600CTRL} was the optical density of the microcultures at 600 nm in synthetic medium without FA (control condition).

2.6. Wort preparation and laboratory-scale microfermentations

A 10 °Plato wort was prepared from dry malt extract (Muntions, United Kingdom). The wort was boiled with 100 g/L Cascade or Talus hops (Yakima Chief Hops, USA) for 10 min, after which the wort was rapidly cooled and stored at 4 °C for 24 h to allow sedimentation. After this period, the clarified wort was carefully decanted into a new container, leaving the sediment behind.

Two different microfermentation methods were employed, differing mainly in final volume, pitching rate, and hop variety, as summarized in Supplementary Table S1.

An inoculation loop was first used to transfer a small amount of yeast to 50 mL YPD medium and propagated under aerobic conditions on a shaker (100 rpm) at 20 °C. After propagation overnight, the cells were washed in sterile Milli-Q-filtered water and re-suspended to achieve a 20% slurry (200 g fresh yeast/L). This slurry was used to inoculate all-malt 10 °P wort. Fermentations were carried out aseptically in Erlenmeyer flasks fitted with glycerol-filled airlocks to allow CO₂ escape while preventing contamination. Fermentations were followed by measuring weight loss two times per day. After 14 days of fermentations, samples were centrifuged, filtered through 0.45 µm pore-size filters and stored at −20 °C for subsequent analyses.

Yeast propagation, fermentation monitoring by CO₂ weight loss, and sample collection procedures were consistent across both methods.

2.7. Physicochemical analyses of beer

Specific gravity, alcohol content, pH, and residual sugar concentration were determined using a DMA 5000 M density meter (Anton Paar GmbH, Austria) equipped with AlcoLyzer Beer ME and pH ME modules. Prior to analysis, samples were transferred to glass tubes and degassed by three successive agitation cycles to eliminate dissolved CO₂. All measurements were conducted on centrifuged, degassed samples to ensure analytical precision, following the procedure of Magalhaes et al. (2016).

Fermentable sugars, ethanol, and glycerol were quantified by high-performance liquid chromatography (HPLC) using a Waters 2695

separation module coupled to a Waters 2414 refractive index detector (Waters Co., Milford, MA, USA). Separation was performed with an Aminex HPX-87H column (300 × 7.8 mm, Bio-Rad) maintained at 55 °C and eluted with 5 mM H₂SO₄ (Titrisol, Merck, Germany) at a flow rate of 0.3 mL min⁻¹.

Quantification of 4-VG was carried out by HPLC-UV, following the protocols of Coghe et al. (2004) and McMurrough et al. (1996). Chromatographic separation was performed using a Thermo Scientific Vanquish system equipped with a C18 column (5 µm, 4.6 × 150 mm). The mobile phase consisted of H₂O/acetonitrile/acetic acid (72:27:1, v/v), with a flow rate of 1 mL min⁻¹. UV detection was carried out across 190–400 nm, with quantification at 260 nm using external calibration curves of 4-VG (0.3–30 mg/L).

2.8. Aroma analysis

Yeast-derived flavor compounds were determined by headspace gas chromatography with flame ionization detector (HS-GC-FID) analysis. Four milliliter samples were filtered (0.45 µm), incubated at 60 °C for 30 min and then 1 mL of gas phase was injected (split mode; 225 °C; split flow of 30 mL min⁻¹) into a gas chromatograph equipped with an FID detector and headspace autosampler (Agilent 7890 Series; Palo Alto, CA, USA). Analytes were separated on a HP-5 capillary column (50 m × 320 µm × 1.05 µm column, Agilent, USA). The carrier gas was helium (constant flow of 1.4 mL min⁻¹). The temperature program was 50 °C for 3 min, 10 °C min⁻¹ to 100 °C, 5 °C min⁻¹ to 140 °C, 15 °C min⁻¹ to 260 °C and then isothermal for 1 min. Compounds were identified by comparison with authentic standards and were quantified using standard curves. 1-Butanol was used as internal standard.

2.9. Sequencing of *PAD1* and *FDC1* genes

To investigate potential loss-of-function mutations in the *PAD1* and *FDC1* genes, genomic PCR amplification followed by Sanger sequencing was performed on FM1 and FM1.45. Yeast genomic DNA (gDNA) was extracted according to Catallo et al. (2021), spectrophotometrically quantified through a Nanodrop, and diluted to 50 ng/µL in ddH₂O. Species-specific primer pairs targeting *FDC1* and *PAD1* genes were designed using Primer 3 and listed in Supplementary Table S2. All PCR reactions were conducted through a T100 Thermal Cycler (Bio-Rad, Hercules, CA, USA) in a final reaction volume of 40 µL containing 100 ng of gDNA, 0.3 µM of each primer, 200 µM of each dNTP, and 0.5 U of DreamTaq DNA Polymerase (Thermo Fisher Scientific, Waltham, MA, USA), according to the manufacturer's instructions. PCR cycling conditions are listed in Supplementary Table S2. A 100 bp Plus DNA ladder (Thermo Fisher Scientific, Waltham, MA, USA) was used as the molecular size marker. The amplification products were visualized by 2% (w/v) agarose gel electrophoresis and purified for Sanger sequencing through the DNA Clean and Concentrator-5 Kit (Cat. No 4013; Zymo Research, Irvine, CA). Sanger sequencing was carried out by Macrogen Inc. (Macrogen Europe, Amsterdam, The Netherlands) with the external primers used for PCR reactions and, in case of *FDC1*, with two additional internal primers (Supplementary Table S2). Sequences were deposited in GenBank database under the accession numbers PX845829 to PX845833.

2.10. Genetic stability of mutagenized strains and ploidy determination

To confirm the strain identity and the genetic stability of the mutagenized strains, species-specific PCR reactions were performed targeting *S. eubayanus* *FSY1* and *S. cerevisiae* *MEX67* genes, respectively. The lager strain JB712 was used as a control. PCR reactions were carried out as reported above and PCR conditions and primers were according to Pengelly and Wheals (2013) and Muir et al. (2011).

Ploidy was assessed by flow cytometry following the protocol of Haase and Reed (2002). Yeast cells were grown overnight in YPD

medium, harvested at 0.1 OD₆₀₀, and washed with 1 mL of 50 mM citrate buffer. Cells were fixed in cold 70% ethanol and incubated overnight at -20 °C. After fixation, they were washed and resuspended in 50 mM citrate buffer containing 0.25 mg mL⁻¹ RNase A and 1 mg mL⁻¹ Proteinase K, and incubated overnight at 37 °C. For DNA staining, cells were incubated with 2 µM SYTOX Green (Life Technologies, USA), and DNA content was analyzed using a FACSaria IIu flow cytometer (Becton-Dickinson). A minimum of 100,000 events were recorded per sample. Ploidy level was estimated by comparing fluorescence intensities with those of diploid (C-6) and tetraploid (YS8(E)) *S. cerevisiae* reference strains (Peter et al., 2018).

Sporulation efficiency and spore viability were evaluated following the protocols described by Catallo et al. (2021).

2.11. Statistical analysis

All assays were carried out in triplicate, except where otherwise indicated. Statistical analyses were performed using one-way ANOVA, unless otherwise specified, with GraphPad Prism v.10 software (GraphPad Software, Inc., San Diego, CA, USA). In all analyses, a *p*-value less than 0.05 was considered statistically significant.

3. Results

3.1. Set-up of UV mutagenesis protocol

UV irradiation is a widely adopted method for inducing random genetic mutations in microorganisms, offering a non-genetically modified approach for improvement of phenotypic traits controlled by one or a poor number of genetic loci, like POF phenotype (Hashimoto et al., 2005; Diderich et al., 2018). To establish an effective UV mutagenesis protocol for obtaining POF⁻ yeast variants, two distinct UV light sources, namely a calibrated crosslinker delivering 5 mJ and an UV lamp placed 50 cm above the plate were compared. Mortality rates were assessed at increasing exposure times for both a diploid strain (BY4243) and a haploid strain (ale11.4B), with the goal of achieving a mortality range between 65 and 85%, considered optimal to balance mutagenic efficiency and the survival of viable clones for screening (Hashimoto et al., 2005). As expected, the haploid strain showed higher sensitivity to UV irradiation, resulting in greater mortality rates compared to the diploid strain under the same conditions (Supplementary Fig. S1). Among the tested settings, UV lamp exposure for 30 s at 50 cm consistently produced mortality levels within the desired range for both strains, identifying this condition as the most suitable compromise between effective mutagenesis and adequate cell viability (Supplementary Fig. S1).

3.2. Constitution of a library of UV mutagenized derivatives of FM1

Ferulic acid (FA) was reported to inhibit POF⁻ yeasts unable to decarboxylate it into 4-VG (Larsson et al., 2001; Fletcher et al., 2019). To generate GMO-free POF⁻ derivatives of the hybrid strain FM1, we applied the previously established UV-based random mutagenesis protocol to exponentially growing FM1 cells. The resulting UV-mutagenized population was screened for the inability to grow on YPDA medium supplemented with 4 mM FA through replica plating. Since the likelihood of obtaining derivatives with the desired phenotype is low through random mutagenesis, we implemented an additional round of random mutagenesis which incorporated a nystatin treatment step. Nystatin is a polyene antifungal compound that binds ergosterol in the cell membrane, forming pores and causing osmotic lysis (Michel, 1977). It preferentially targets rapidly dividing cells, thereby enriching for slow-growing mutants potentially carrying beneficial mutations (Colomer, 2019; Snow, 1966; Strömnaes and Mortimer, 1968). From the standard and nystatin-assisted mutagenesis experiments 18 and 6 FA-sensitive colonies were recovered, corresponding to isolation efficiencies of

13.3% and 0.4%, respectively.

3.3. Screening for POF^- phenotype using FACI and CASI metrics

Diderich et al. (2018) demonstrated that the POF^- trait can be validated through the FACI and CASI metrics. FACI reflects the ability of a strain to convert FA into 4-VG, while CASI measures growth inhibition in the presence of CA. High FACI and low CASI values are characteristic of a POF^+ phenotype (Diderich et al., 2018).

FACI and CASI values were determined for all mutants as well as for their parental hybrid strain FM1. Consistent with its POF^+ phenotype, the FM1 hybrid exhibited a high FACI value (95%) (Fig. 1A). Most mutants showed FA-to-4-VG conversion rates between 45% and 100%. Two mutants (FM1.21 and FM1.53) exhibited a significant increase in FACI relative to FM1, one mutant was comparable to FM1, and the remaining 21 mutants exhibited significantly reduced FACI values (Fig. 1A). Among these, the UV-mutagenized derivative FM1.45 displayed the lowest FACI value, followed by the nystatin-treated

derivative FM1.6N (Fig. 1A). These markedly reduced FACI values suggest a complete loss, or at least a substantial reduction of decarboxylation activity in FM1.45 and FM1.6N.

CASI analysis revealed a diversity of phenotypes among the UV-derived isolates (Fig. 1B). Eight derivatives appeared less sensitive to CA than FM1 ($p < 0.05$). However, comparison of OD_{600} values under control and treated conditions indicated that these strains grew poorly even under control conditions, resulting in artificially low CASI values. We hypothesized that untargeted mutagenesis impaired their growth ability. Nine derivatives exhibited CASI values statistically comparable to FM1, including the candidate FM1.6N (Fig. 1B). This finding led us to discard FM1.6N as potential POF^- derivative of FM1, despite its initial selection based on FACI. The remaining 7 derivatives exhibited CASI values statistically higher than FM1 (Fig. 1B). Among these, FM1.45 was the only derivative exhibiting both a statistically higher CASI value and statistically lower FACI value compared with FM1 ($p < 0.05$; Fig. 1B). Taking together, these results identify FM1.45 as the most promising POF^- mutant, showing both significant reduction in FACI and a

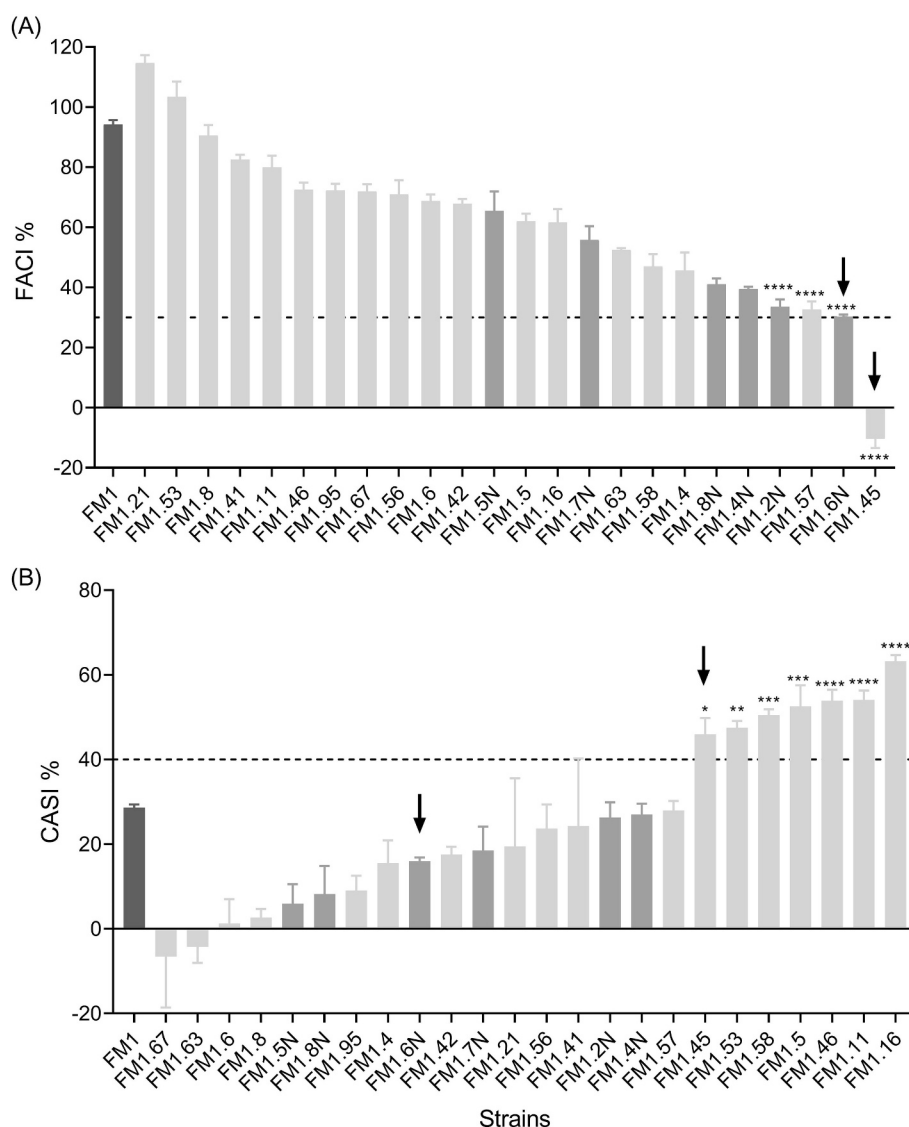


Fig. 1. Comparison of FACI (A) and CASI (B) indicators between the original hybrid strain FM1 and the 24 isolates generated by UV mutagenesis. Mutants with N suffix are treated with nystatin and are reported in dark grey. All values are expressed as percentage and represent the mean of at least three independent replicates. Bars, when visible, represent standard deviation. Horizontal dotted black line represents FACI reduction of 60% respect to FM1 (corresponding to value percentage of app. 30%) in panel A and CASI increase of 70% respect to FM1 (corresponding to a value percentage of app. 40%) in panel B, respectively. For brevity, significant differences were reported only for FACI values below 30% and for CASI values above 40%, as follows: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$. Graphs were plotted with GraphPad version 8 (San Diego, CA, USA).

significant increase in CASI relative to the parental strain FM1.

3.4. Validation of FM1.45 derivative as POF⁻ candidate

The POF⁻ phenotype of the UV mutagenized derivative FM1.45 was further confirmed by quantifying the residual FA and the produced 4-VG in supernatants obtained from FACS assay. Specifically, supernatants of FM1 and FM1.45 were analyzed by high-resolution mass spectrometry. The analysis revealed the presence of both FA and its isomer, isoferulic acid, in the non-inoculated medium, although at low concentrations. The presence of isoferulic acid may result from chemical isomerization (e.g., due to heat or alkaline pH), or from impurities in commercial ferulic acid preparation. FM1 exhibited a degradation percentage of 57.55%, which was significantly higher than that of FM1.45. In contrast, FM1.45 retained a significantly higher concentration of both ferulic and isoferulic acid in the supernatant compared to the original hybrid FM1, indicating a lower decarboxylation capacity and confirming its reduction in POF phenotype (Table 2).

POF⁻ derivatives are expected to exhibit reduced growth on FA-containing medium due to their inability to convert the weak acid FA into 4-VG. To evaluate growth differences in the presence of FA, parental hybrid strain FM1 and its POF⁻ derivative FM1.45 were cultured in both control synthetic medium and in synthetic medium supplemented with 4 mM FA. Strain A-6305 was used as POF⁻ control. In the control medium, all strains showed similar growth dynamics, reaching stationary phase after approximately 21 h (Supplementary Fig. S2A). This result suggests that UV mutagenesis did not impair growth ability in the derivative FM1.45. Conversely, in the medium containing 4 mM FA, a significant growth difference was observed, with FM1 showing higher growth than FM1.45 (Supplementary Fig. S2B). To quantitatively assess growth inhibition induced by FA, the percentage of growth inhibition was calculated by comparing growth in FA-supplemented medium to control medium at 8 h, 10 h and 12 h (Fig. 2). Data confirmed that FA inhibited FM1.45 more effectively than FM1 (Fig. 2A). The impaired growth of FM1.45 in the presence of FA is consistent with the reduced ability of FM1.45 to convert FA into 4-VG.

3.5. Characterization of the POF⁻ derivative FM1.45 in wort fermentation

The fermentative performance of the parental strain FM1 and its potential POF⁻ derivative FM1.45 was evaluated under four different combinations of fermentation temperatures (10 °C and 20 °C) and pitching rates (0.125% and 0.5%). Fermentation kinetics were monitored by measuring weight loss over time, which reflects CO₂ release and thus fermentation progress. As expected, an increased pitching rate (0.5%) resulted in faster fermentation kinetics and overall improved fermentative performance compared to the lower pitching rate (0.125%).

Table 2

Residual concentrations and degradation percentages of ferulic acid and isoferulic acid by the hybrid strain FM1 and its UV-mutagenized derivative FM1.45 after growth in YPD supplemented with 1 mM ferulic acid. Values are shown as mean ± SD (n = 3). Different superscript letters indicate significant differences between FM1 and FM1.45 within the same parameter (two-tailed unpaired t-test, p < 0.05). NA, not applicable.

Strain	Compound	Residual concentration (mM)	Degradation (%)	Total degradation (%)
FM1	Ferulic acid	0.39 ± 0.01 ^a	57.61 ± 1.00 ^a	59.00 ± 0.71 ^a
FM1	Isoferulic acid	0.04 ± 0.00 ^a	50.00 ± 2.43 ^a	–
FM1.45	Ferulic acid	0.70 ± 0.02 ^b	23.91 ± 2.00 ^b	28.00 ± 2.12 ^b
FM1.45	Isoferulic acid	0.02 ± 0.00 ^b	75.00 ± 0.13 ^b	–

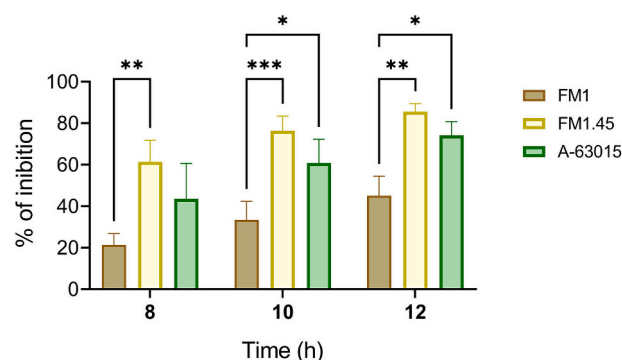


Fig. 2. Growth inhibition (%) of the hybrid strain FM1 and its UV-mutagenized derivative FM1.45 evaluated after 8, 10 and 12 h of exposition to 4 mM FA. Yeast growth was measured over time as optical density at 600 nm (OD₆₀₀). Each condition was tested in three independent biological replicates. The POF⁻ strain A-63015 served as a control. Significant differences were evaluated by one-way ANOVA (p < 0.05) relative to FM1 and reported as follows: *, p < 0.05; **, p < 0.01; ***, p < 0.001; ****, p < 0.0001. Graphs were plotted with GraphPad version 8 (San Diego, CA, USA).

At 20 °C with a low pitching rate (0.125%), the mutant strain FM1.45 displayed the highest fermentation activity among all tested strains (Fig. 3A). At 10 °C with a low pitching rate, the hybrid FM1 and the mutant FM1.45 showed comparable fermentation profiles, both outperforming the parental strains (Fig. 3B). At 20 °C with a high pitching rate (0.5%), the *S. cerevisiae* strain Y15.2B exhibited the highest fermentation performance (Fig. 3C). Finally, at 10 °C with a high pitching rate, the hybrid FM1 and the mutant FM1.45 again showed similar performance, surpassing the parental strains (Fig. 3D).

At the end of fermentation, physicochemical properties of the final beers and 4-VG content were determined. Density, pH, °Plato, °Brix and apparent degree of fermentation (ADF%) were consistent with the fermentation trends and are provided in Supplementary Table S3.

Ethanol, maltose, and maltotriose quantification by HPLC revealed that, at a pitching rate of 0.125%, notable differences in sugar (maltose and maltotriose) consumption were observed, with more efficient fermentation at 20 °C than at 10 °C (Supplementary Fig. S3). For maltose, the most pronounced differences were observed for FM1, FM1.45, and the *S. eubayanus* x *S. uvarum* parental strain NBRC1948, which fermented maltose more efficiently at 20 °C than 10 °C (p < 0.05). At 10 °C, the maltose consumption of FM1 and its derivative FM1.45 more closely resembled that of the parental strain Y15.2B than that of NBRC1948. Regarding maltotriose, the derivative FM1.45 and the parental strain NBRC1948 were less efficient in maltotriose consumption at 10 °C relative to 20 °C (p < 0.05) (Supplementary Fig. S3).

At the pitching rate of 0.5%, sugar consumption patterns appeared more consistent and linear across strains (Supplementary Fig. S4). No significant differences in ethanol production were across strains and between conditions (10 vs. 20 °C). As expected, the *S. cerevisiae* parental strain Y15.2B consumed significantly less maltose and maltotriose at 10 °C than at 20 °C, whereas the *S. eubayanus* x *S. uvarum* parental strain NBRC1948 did not exhibit significant differences in maltose and maltotriose consumption between the two temperatures. Notably, NBRC1948 completely consumed maltotriose at both temperatures, confirming previous findings reported by Catallo et al. (2021). The POF⁻ mutant of FM1, FM1.45, showed lower maltotriose consumption than the original hybrid strain, but higher consumption than the parental strain Y15.2B at 10 °C, whereas completely consumed maltotriose at 20 °C, similarly to FM1 (Supplementary Fig. S4).

HPLC determination of 4-VG was carried out in fermented wort samples obtained from microfermentation trials at 10 °C and 20 °C with the lowest pitching rate. FM1 and its parental strain *S. cerevisiae* Y15.2B exhibited the highest production of 4-VG at both the temperature, while

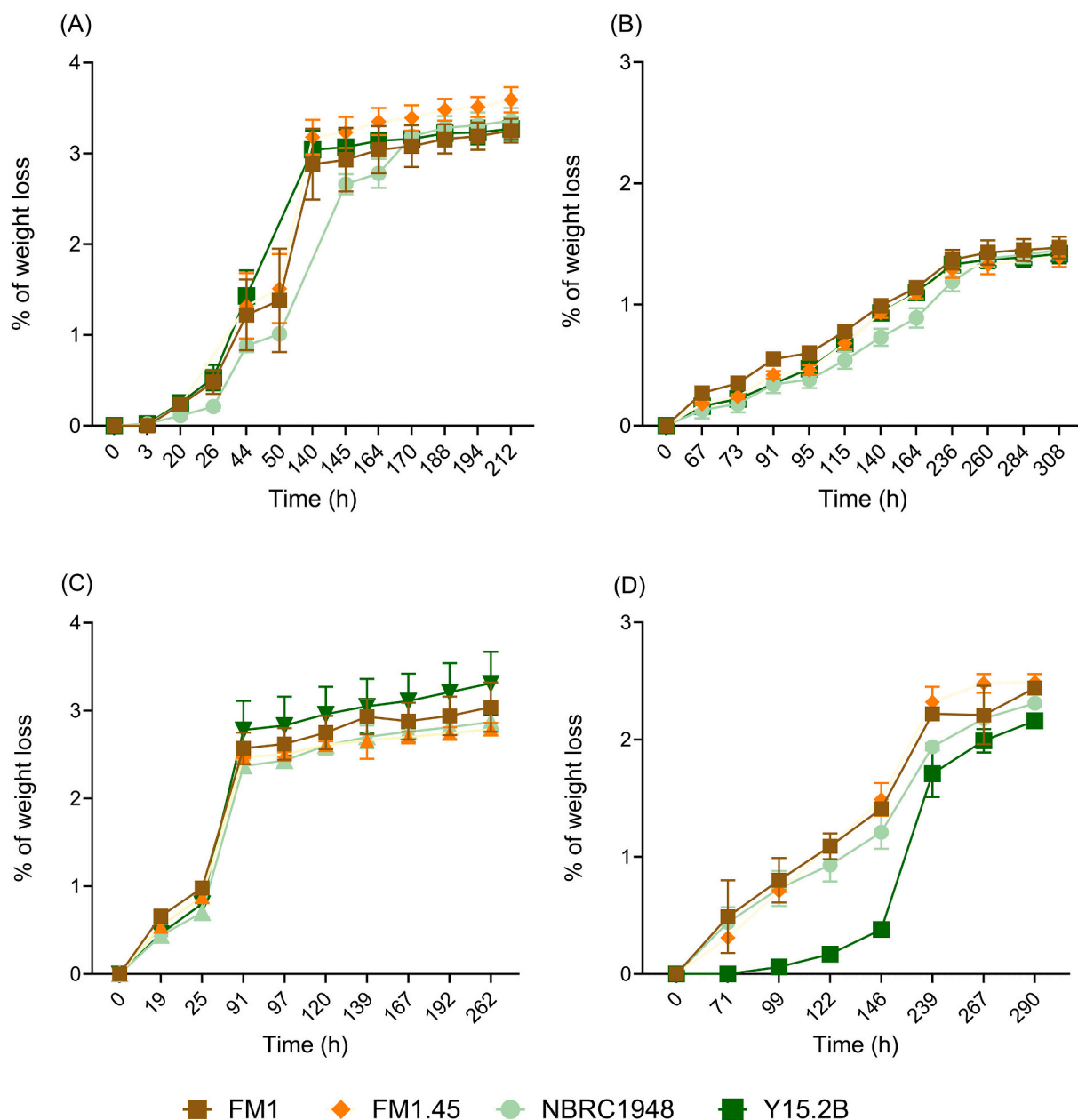


Fig. 3. Fermentation kinetics of the hybrid strain FM1 and its POF⁻ UV-mutagenized derivative FM1.45. The following combinations of temperature and pitching rate were considered: A) 20 °C, 0.125% pitching rate; B) 10 °C, 0.125% pitching rate; C) 20 °C, 0.5% pitching rate; D) 10 °C, 0.5% pitching rate. The parental strains of FM1, namely *S. cerevisiae* Y15.2B and *S. eubayanus* x *S. uvarum* NBRC1948, were included for comparative purposes. CO₂ loss was monitored as a proxy for fermentation rate during microfermentations in 10°P wort. Each curve represents the mean weight loss (%) of three independent replicates; error bars indicate standard deviation.

the cryotolerant parental strain NBRC1948 significantly reduced the 4-VG production at 10 °C compared with 20 °C (Fig. 4). The mutant strain FM1.45 exhibited a significant reduction in 4-VG production at both the temperatures compared with the hybrid FM1 and *S. cerevisiae* Y15.2B (Fig. 4). At 10 °C FM1.45 was the lowest producer of 4-VG even if not statistically different from the parental strain *S. eubayanus* x *S. uvarum* NBRC1948 (Fig. 4B).

Potential differences in aromatic compound production were also assessed in wort fermented by strains FM1 and FM1.45 at 10 °C and 20 °C (Supplementary Table S4). Overall, both strains exhibited comparable concentrations of the main aroma-active compounds, indicating that the mutagenesis process did not markedly affect the overall aromatic balance. Minor variations observed between replicates contributed to the variability in compound concentrations but did not alter the

general trend. Acetaldehyde levels were notably higher at 10 °C, exceeding the reported sensory threshold of 25 µg/L. This may contribute to light green or fruity notes, which are often associated with low-temperature fermentation. Ethyl acetate and higher alcohols (propanol, isobutanol, isoamyl alcohol, and 2-methylbutanol) showed a temperature-dependent trend, with higher concentrations detected at 10 °C than at 20 °C. These results are consistent with reduced volatilization and slower metabolic turnover at low fermentation temperatures.

When comparing measured concentrations to sensory thresholds, acetaldehyde, ethyl acetate, and isoamyl alcohol were above the olfactory detection limit in at least one condition. Overall, FM1.45 exhibited an aroma profile closely aligned with that of the parental strain FM1, confirming that the UV mutagenesis did not substantially alter the production of key volatile aroma compounds (Fig. 5). Differences in

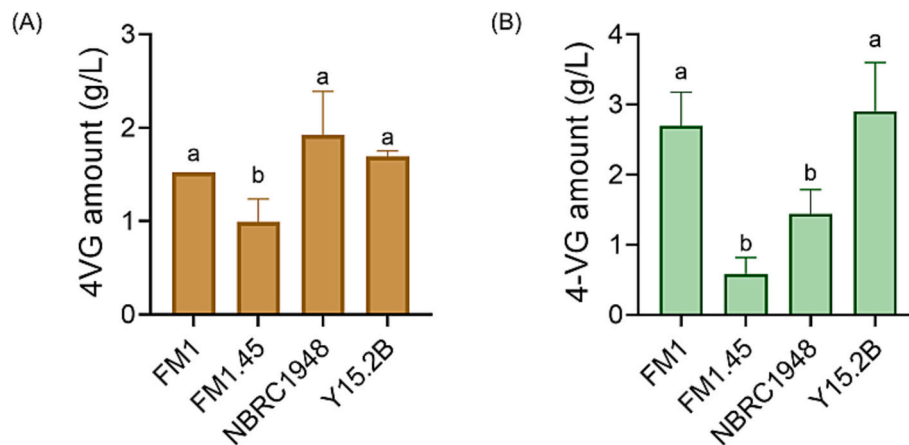


Fig. 4. 4-VG quantification in beer samples fermented by the hybrid strain FM1 and its UV-mutagenized derivative FM1.45 (pitching rate 0.125%) at 20 °C (A) and 10 °C (B). The data represents the mean of at least three independent replicates and bars represent standard deviation. Statistical analysis was carried out through one-way ANOVA and different letters indicate significant differences among strains ($p < 0.05$). Graphs were plotted with GraphPad version 8 (San Diego, CA, USA).

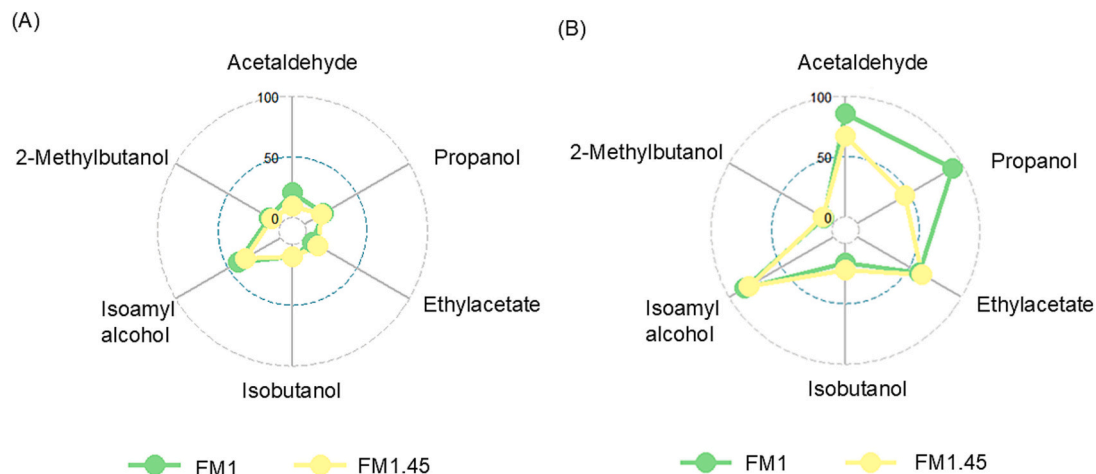


Fig. 5. Radar charts comparing the aromatic compound profiles of strains FM1 and FM1.45 at 20 °C (A) and 10 °C (B). Values represent the concentration of volatile compounds expressed in mg/L; maximum reference value of 100 mg/L for visualization. Each polygon corresponds to one strain, illustrating differences in higher alcohol and ester production between strains.

compound concentrations can also be attributed to replicate variability, which is expected in fermentative processes.

3.6. Sequencing of *PAD1* and *FDC1* genes in the *POF*[−] derivative FM1.45

The *PAD1* and *FDC1* genes of the mutagenized strain FM1.45 and its parents FM1, NBRC1948, and Y15.2B were amplified with *S. cerevisiae*- and *S. eubayanus*/*S. uvarum*-specific primer pairs and submitted to Sanger sequencing. Amplification of the *S. cerevisiae* copies produced the expected fragments of ~900 bp for *PAD1* and ~1750 bp for *FDC1* in FM1, FM1.45 and in the *S. cerevisiae* parental strain Y15.2B (Fig. 6). In contrast, while PCR amplification of *PAD1* and *FDC1* from the *S. eubayanus* sub-genome succeeded in FM1 and NBRC1948, it failed in Y15.2B and FM1.45 (Fig. 6). This suggests that right telomeric portion of chromosome XIII, carrying the *PAD1* and *FDC1* genes in *S. eubayanus* and *S. uvarum*, from the NBRC1948 sub-genome may be deleted or substantially rearranged in FM1.45.

Sequencing of *S. cerevisiae* *FDC1* alleles from Y15.2B, FM1, and FM1.45 revealed three polymorphisms compared with the S288C reference genome, one of them leading to a nonsynonymous mutation (Asp55Tyr). In addition, FM1.45 exhibited a frameshift mutation at the position 390 resulting in the introduction of a premature stop codon at position 425. As a result, the predicted Fdc1 protein of FM1.45 was

truncated compared with those of Y15.2B and FM1 (Fig. 6E). We speculated that this mutation contributes to a loss-of-function responsible for the lower production of 4-VG detected in FM1.45 compared with FM1.

In the case of *PAD1*, no *de novo* mutations were detected in *PAD1* gene from FM1.45 compared with *S. cerevisiae* Y15.2B, except for the substitution Gly220Asp (Supplementary Fig. S5).

3.7. Confirmation of hybrid nature, reproductive capacity, ploidy determination

To confirm the hybrid nature of the *POF*[−] selected variant FM1.45, species-specific PCR assays of housekeeping genes were performed using primer pairs targeting *S. cerevisiae* *MEX67* gene and *S. eubayanus* *FSY1* gene, respectively. Positive amplifications were observed in FM1.45 and the original FM1 hybrid with both primer pairs, confirming that both hybrid sub-genomes remained present in FM1.45 after mutagenesis (Fig. 7A and B).

Since eukaryotic species are reproductively isolated by sterility barriers that prevent interspecies fertilization, typically interspecies hybrids are expected to be sterile due to the inability either to sporulate or to produce viable spore (Delneri et al., 2003). Congruently, hybrid FM1 exhibited a sporulation efficiency of 75.7% and the resulting spores were unviable (Fig. 7C and E). In contrast, the mutant strain FM1.45

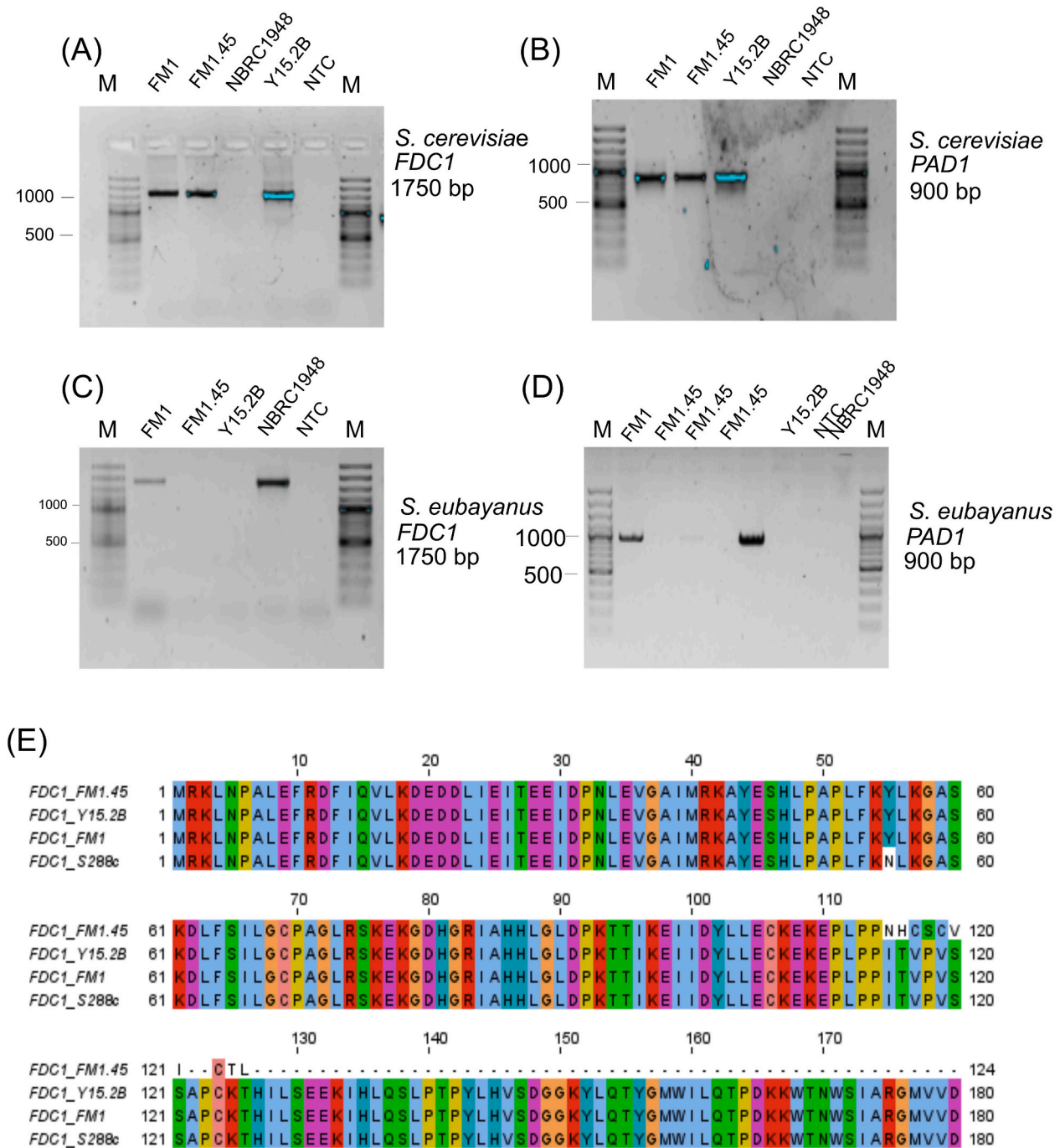


Fig. 6. Analysis of *PAD1* and *FDC1* genes in the POF⁻ derivative FM1.45. (A) Electrophoretic run on 2% (w/v) agarose gel of the PCR amplicons corresponding to the *FDC1* gene for the allelic variant of *S. cerevisiae*. (B) PCR amplification of the *PAD1* gene in *S. cerevisiae*. (C) PCR amplification of the *FDC1* gene in *S. eubayanus*. (D) PCR amplification of the *PAD1* gene in *S. eubayanus*. M indicates the GeneRuler 100 bp Plus DNA Ladder (Thermo Scientific, Waltham, USA). (E) Multiple amino acid sequences alignment of the *S. cerevisiae* Fdc1 proteins from the mutagenized strain FM1.45 and its parental strains FM1 and *S. cerevisiae* Y15.2B. Fdc1 from *S. cerevisiae* strain S288C was used as reference. The alignment was visualized by Jalview (Waterhouse et al., 2009) and only the first 180 aa were shown for brevity.

exhibited a reduced sporulation efficiency (24.5%) compared with the parent FM1, but a spore viability of 32.7% (Fig. 7D and E). The viability of F1 meiotic segregants of FM1.45 indicated that, unlike its parent FM1, the derivative FM1.45 could be fertile.

To investigate the basis of offspring viability in FM1.45, ploidy content was determined in FM1, FM1.45, and in the parental strains Y15.2B and NBRC1948 through flow cytometry (Fig. 7E). The main peak of FM1 falls between the diploid and tetraploid controls, consistent with ~3n DNA content, whereas FM1.45 displays a peak overlapping the tetraploid control, indicating an increase in DNA content consistent with

~4n (Supplementary Fig. S6). The parental strains displayed DNA contents consistent with tetraploidy for *S. cerevisiae* Y15.2B and diploidy for *S. eubayanus* × *S. uvarum* NBRC1948, in agreement with the intermediate DNA-content profile observed for FM1 (Fig. 7E). Overall, these data suggest that UV mutagenesis was associated with an increase in genome DNA content in FM1.45, which may have contributed to the partial restoration of spore viability. However, as flow cytometry measures total DNA content, aneuploidy or segmental duplications cannot be excluded without karyotype- or sequencing-based analyses.

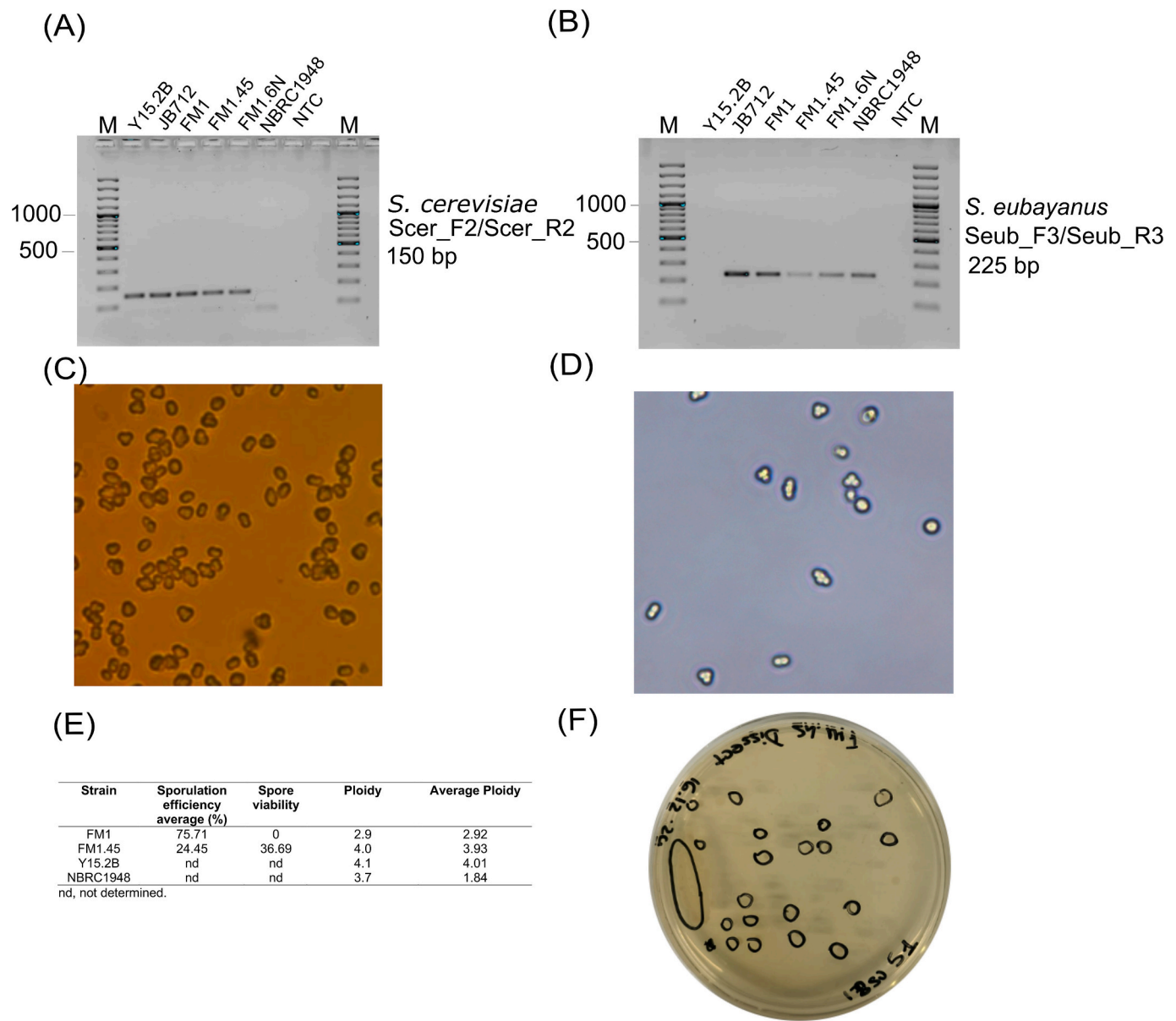


Fig. 7. Hybrid identity confirmation, fertility assays, and ploidy determination. (A) Electrophoretic run on 2% (w/v) agarose gel of the PCR amplicons obtained with *S. cerevisiae*-specific primers. M indicates the GeneRuler 100 bp DNA Ladder (Thermo Scientific, Waltham, USA). (B) Electrophoretic run on 2% (w/v) agarose gel of the PCR amplicons obtained with *S. eubayanus*-specific primers. M indicates the GeneRuler 100 bp DNA Ladder (Thermo Scientific, Waltham, USA). (C) Optical microscopy image (40×) of FM1 sporulation assay. (D) Optical microscopy image (40×) of FM1.45 sporulation assay. (E) Table summarizing sporulation percentage, spore viability, and ploidy values assessed by flow cytometry for FM1 and FM1.45. In flow cytometry diploid strain C-6 and tetraploid strain YS8(E) were used as references. (F) Plate showing ascus dissection and monosporic culture isolation from FM1.45.

4. Discussion

Hybridization and UV mutagenesis are conventional methods widely used to improve industrial yeast strains and have often been applied independently for strain development (Nasuti and Solieri, 2024). While hybridization is particularly effective for enhancing or combining in a single strain complex, multigenic traits, UV mutagenesis is preferable when targeting traits are governed by a limited number of genetic loci (Giudici et al., 2005). Several successful attempts have been made to generate *de novo* lager strains through *in vitro* hybridization, aimed at mimicking the characteristics of traditional lager yeasts (Steensels et al., 2014; Hebly et al., 2015; Magalhães et al., 2017; Krogerus et al., 2017). However, the use of parental lineages lacking desired domestication signatures has often resulted in *de novo* hybrids with limited scalability at the industrial level.

Among the domestication signatures, the POF^- phenotype is strictly related to the inability to produce 4-VG. Compared to other classes of off-flavor molecules, 4-VG exhibits a lower sensory threshold value and confers distinctive clove-like and phenolic aroma that is highly undesired in lager beers (Vanbeneden et al., 2007). Furthermore, even low concentrations of 4-VG can modulate or mask the perception of other flavors, thereby limiting the development of beverages with innovative and diverse aroma profiles (Gallone et al., 2016). Consequently, recent efforts have focused on generating POF^- *de novo* lager hybrids, by targeting the inability to produce 4-VG (Diderich et al., 2018; Mertens et al., 2019). In creating POF^- lager hybrids, these studies typically prioritized the construction of POF^- parental strains, followed by the marker-assisted development of *S. cerevisiae* × *S. eubayanus* hybrids. In all cases, GMO-based techniques were employed either to disrupt the *PAD1* and *FDC1* gene clusters in the parental strains or to facilitate

hybrid selection and optimization.

In the present study, we developed a fully GMO-free strategy consisting of two steps: (i) hybrid strain construction without any marker-assisted selection, and (ii) elimination of *PAD1* and *FDC1* function through UV mutagenesis. We demonstrated that UV mutagenesis applied after interspecies hybridization represents an effective, non-GMO strategy which exploits hybrid genome instability to effectively improve brewing-relevant traits while maintaining strain performance. Hybrid construction involved a *S. cerevisiae* parental strain originating from a non-brewing niche, with the aim to expand the diversity of beer aroma profiles. Sourdough yeast communities represent a valuable reservoir of novel *S. cerevisiae* strains able to ferment brewer's wort (Marongiu et al., 2015; Mascia et al., 2015; Rossi et al., 2018; Cubillos et al., 2019; Catallo et al., 2020; Sánchez-Adriá et al., 2024) and are typically used to produce Finland's sahti beer (Ekberg et al., 2015). In our previous work, mating of a monospore culture of a *S. cerevisiae* sourdough strain with the cold tolerant *S. eubayanus* x *S. uvarum* strain NBRC1948, isolated from brewing environment, resulted in an interspecies hybrid displaying wort fermentation performance superior to that of the parental strains. The cold-tolerant strain NBRC1948 exhibited the ability to consume maltotriose, making it suitable alternative to maltotriose negative *S. eubayanus* strains as a parental lineage in sexual breeding programs (Catallo et al., 2021). This trait is likely related to a *S. cerevisiae*-introgressed DNA region containing the maltose transporter gene *MAL31* and the maltose and maltotriose transporter gene (*MTY1/MTT1*). The resulting hybrid FM1 likely inherited the NBRC 1948 haplotype carrying this introgressed region, leading to higher sugar consumption compared with the parental strains.

UV exposition induces DNA lesions which trigger a hypermutable state during DNA repair of hypermutability, thereby promoting the accumulation of multiple mutations (Yang et al., 2008). In this study, induction of DNA damage through UV exposition was performed with and without nystatin followed by negative selection in the presence of FA. The objective was to isolate derivatives unable to grow in the presence of FA due to spontaneous nonsense mutations or gene loss of *PAD1* or *FDC1*, the two genes responsible for the POF^+ phenotype and involved in FA detoxification. As expected, the mutagenesis of the hybrid strain FM1 in combination with nystatin resulted in a lower colony yield compared to untreated controls. This reduced growth is likely due to improved selection efficiency and the suppression of fast-growing cells that may not have undergone mutagenesis, thereby decreasing the number of non-mutated, false-positive colonies, as previously reported (Strömnaes and Mortimer, 1968). However, in our experimental setup, conventional UV mutagenesis without nystatin proved to be the most effective strategy, allowing the isolation of FM1.45 as most promising POF^- derivative of FM1.

According to Diderich et al. (2018), FACI assay quantitatively measures the reduction in ferulic acid decarboxylation ability, while the CASI assay serves both to exclude strains with impaired fitness and to further confirm sensitivity to weak acid stress, which may reflect compromised detoxification via decarboxylation. This bimodal screening approach identified FM1.45 as the most promising POF^- derivative of hybrid strain FM1, whereas the FM1.6N candidate was classified as false positive. Survival analysis in the presence of FA and FA determination after FACI assay further corroborated the POF^- phenotype of FM1.45 relative to its parental strain FM1.

A major limitation in untargeted random mutagenesis methods, like UV mutagenesis, is the potential introduction of accidental mutations that may affect industrially relevant traits unrelated to the selected phenotype. Wort microfermentation experiments indicated that UV mutagenesis did not significantly impair fermentative performance or alter the overall aroma profile of FM1.45 compared with the parental hybrid FM1. At 20 °C and high pitching rate (0.5%), Y15.2B outperformed all strains, consistent with optimal growth conditions and rapid sugar metabolism, whereas FM1 and FM1.45 retained superior fermentation efficiency under suboptimal low-temperature conditions.

The main difference between FM1 and FM1.45 was the significant reduced 4-VG production in wort fermented by FM1.45 compared with FM1 and the parental strain *S. cerevisiae* Y15.2B. Under these conditions, FM1.45 also exhibited a slight reduction in maltotriose consumption compared with FM1. Mutation-related alterations in sugar transport or utilization pathways cannot be excluded and may be clarified by future whole genome sequencing. Analysis of 4-VG production in wort fermented by the parental strains indicated that *S. cerevisiae* Y15.2B was the parental lineage most contributing to POF^+ phenotype of FM1 especially at 10 °C, although *S. eubayanus* x *S. uvarum* NBRC1948 also produced significant amount of 4-VG at 20 °C. Overall, these results support the POF^- phenotype of FM1.45 under controlled, laboratory-scale microfermentation conditions relevant to lager brewing. For industrial validation, the next steps will include pilot-scale fermentation and sensory evaluation of the final beer.

These observations suggested that the reduced POF phenotype detected in FM1.45 could involve genetic changes affecting *PAD1* and *FDC1* genes in both the *S. cerevisiae* and *S. eubayanus* sub-genomes. Consistently, species-specific PCR and sequencing analyses showed that FM1 yielded amplicons for both parental *PAD1* and *FDC1* copies, whereas in FM1.45 only the *S. cerevisiae* amplicon was detected. This pattern is consistent with deletion or major rearrangement of the targeted *S. eubayanus* locus, although primer-site alterations cannot be excluded without genome-wide analysis. In line with previous reports that UV mutagenesis can generate point mutations as well as larger-scale genomic changes (e.g., structural variants, copy-number variation, loss of heterozygosity, and aneuploidies) (Zhu et al., 2025), we detected in FM1.45 a frameshift mutation in the remaining *S. cerevisiae* *FDC1* allele. Consistent with known UV mutational signatures (Brash, 2015), a thymine insertion within a cytosine-containing dipyrimidine sequence introduced a premature stop codon, predicting a truncated *FDC1* protein and likely contributing to the reduced 4-VG formation observed in wort fermentations. Notably, FM1.45 still displayed partial ferulic depletion in YPD medium supplemented with FA (Table 2). This is consistent with reports that *PAD1/FDC1* genotype alone does not always fully predict the magnitude of the POF phenotype across brewing strains and suggests that additional strain-specific factors and/or non-decarboxylative processes may contribute to phenolic acid disappearance (Gonçalves et al., 2016; Kerruish et al., 2024). Importantly, the key brewing-relevant readout, the 4-VG formation in wort, was markedly reduced in FM1.45.

Finally, UV exposure was associated with an increase in nuclear DNA content in FM1.45 relative to FM1, consistent with a shift from $\sim 3n$ toward $\sim 4n$ (Fig. 7E; Supplementary Fig. S6). This observation is compatible with UV-induced genomic instability in hybrids, that may display elevated genome instability compared with homozygous lineages (Dutta et al., 2017). A possible mechanism underlying the increased DNA content is whole-genome duplication via endoreduplication, which has been reported to restore fertility in interspecific yeast hybrids (Pfliegler et al., 2012; Charron et al., 2019); however, other explanations (e.g., aneuploidy or segmental duplications) cannot be excluded with the methods used here. Importantly, FM1.45 showed measurable spore viability, in contrast to the sterile FM1, indicating that the UV-derived variant can generate viable gametes and potentially serve as a fertile starting point for further breeding. In conclusion, this study demonstrates that the sequential combination of interspecies hybridization and UV mutagenesis represents an effective, fully GMO-free strategy to generate POF^- lager yeast hybrids with robust fermentation performance, and can be accompanied by changes in subgenomic contributions and genome copy number, thereby expanding the toolbox for sustainable industrial yeast improvement.

CRedit authorship contribution statement

Chiara Nasuti: Writing – review & editing, Writing – original draft, Investigation, Data curation. **Gabriele Iannuzziello:** Investigation. **Viola Ceramelli Papiani:** Investigation. **Davide Tagliacuzzi:** Writing

– review & editing, Investigation, Data curation. **Kristoffer Krogerus:** Writing – review & editing, Writing – original draft, Methodology, Data curation, Conceptualization. **Lisa Solieri:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

CN received partial funding from AEB S.p.A.; however, this funding did not influence the study design, data collection, analysis, interpretation of results, or the decision to publish.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijfoodmicro.2026.111786>.

Data availability

Data will be made available on request.

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